



BUILD SPECIFICATION

How to Build It. Properly.

The engineering specification for a WhatsApp AI rental agent — features, flows, tools, data model and build order. No market research, no filler.

DOCUMENT

Build spec v1

DATE

1 September 2026

MVP ESTIMATE

6–8 weeks

AUDIENCE

Engineering + product

01 The five laws

Nothing else in this document matters if these break. Each one is enforced in **code**, not in the prompt — a prompt is a request, a constraint is a guarantee.

LAW 1

The AI never asserts availability

It may only repeat what `check_availability()` returned *in the same turn*. Not from memory, not from earlier in the conversation, not from the catalogue.

Enforce: the model cannot emit an availability statement without a tool result in that turn's context. Booking re-validates atomically and fails closed.

LAW 2

The AI never produces a number about money

No totals, no VAT, no per-day maths, no "roughly around". Every figure is a field on the quote object returned by `get_quote()`. The model formats; it does not calculate.

Enforce: output filter rejects currency figures not present in the active quote payload.

LAW 3

"Confirmed" is a database state, not a sentence

The word only appears after `create_booking()` returns a booking ID. Before that: "let me lock that in".

Enforce: the confirmation message is a template rendered from the booking record, not free text.

LAW 4

Payment is confirmed by the provider, never by the customer

"I paid" / "sent it" / a screenshot of a transfer are not evidence. Only a payment-provider webhook moves the booking forward.

Enforce: booking state machine has no transition from `awaiting_payment` triggered by a message.

LAW 5

Policy comes from the knowledge base, in approved translations

The model never composes or translates insurance, liability, deposit or fine wording. It selects an approved chunk. If there is no chunk, it says it will get a definitive answer and escalates.

Enforce: policy answers are retrieved by ID from a versioned KB, per language, human-approved.

WHY THESE FIVE AND NOT "BE HELPFUL AND ACCURATE"

Every one of them maps to a way this business actually loses money: a double-booked car, an invoice that doesn't match the quote, a car dispatched against an unpaid booking, and a promise about insurance the company then has to honour. Politeness is a preference. These are the product.

02 Feature spec

Ordered by what the customer actually needs, not by what is fun to build. **P0** ship or don't launch · **P1** ship within 8 weeks of launch · **P2** later. Autonomy: **A** AI alone · **B** AI + backend validation · **C** human approves · **D** human only.

Before the booking

PRI	AUT	FEATURE	WHAT IT MUST DO
P0	A	Instant first reply	Acknowledge and start qualifying in <10 seconds, 24/7, in the customer's language. Never "we'll get back to you".
P0	A	Partial-message understanding	Extract whatever slots exist from one free-form line ("Corolla tomorrow 4 days Marina"), ask only for what's missing, max two at a time.
P0	A	Eligibility check	Resident vs tourist vs GCC → correct document set. Age vs vehicle class. Licence country → is an IDP needed. Tell them <i>before</i> they get attached to a car.
P0	B	Live availability	Real unit-level answer. If unavailable, return 2–3 verified alternatives with prices in the same message — never a dead end.
P0	B	Binding itemised quote	Line items, VAT, deposit type and amount, mileage cap and overage rate, what is <i>not</i> included, quote ID, expiry time. Same object later becomes the invoice.
P0	B	Soft hold	30-minute vehicle hold on quote issue, expiry stated in the message, released automatically and the customer told.
P0	A	Terms recap	Five lines before confirmation: deposit, mileage, exclusions, fuel, cancellation. This is what kills disputes later.
P0	A	Car photos & specs	Send images from the catalogue with the quote. Customers buy the picture.
P1	B	Price-anchor reconciliation	Handle "I saw AED 89/day on OneClickDrive" — state the live price, explain the difference, offer the nearest match at that price if one exists.
P1	A	Budget-led recommendation	"Cheapest that fits 5 people and luggage" → 3 options, sorted, with the trade-off named.
P2	B	Waitlist	Notify when a wanted-but-unavailable car frees up.

Closing the booking

PRI	AUT	FEATURE	WHAT IT MUST DO
P0	B	In-chat document capture	Accept photos, pull them out of WhatsApp immediately, classify type, detect blur/expiry, ask for a retake in one line if unusable.
P0	C	Document verification	OCR pre-checks (name match, expiry, licence class) then human approve/reject with a reason. Never AI-rejected at MVP.
P0	B	Payment link	Provider-generated link in-thread. Supports full prepay, deposit-only pre-auth, or pay-on-delivery.
P0	B	Atomic booking	Re-validate, write, return ID. On conflict: apologise, re-offer, never silently confirm.
P0	B	Confirmation artefact	Template message + PDF: booking ID, car, plate, times, address, itemised price, deposit terms, policy. Generated from the record.
P1	B	Pay-now incentive	Small discount for paying immediately vs on delivery. Converts a chat quote into cash today.
P1	B	Returning-customer fast path	Documents already on file and unexpired → skip straight from quote to payment. One-tap rebooking of the last car.

Delivery & handover

PRI	AUT	FEATURE	WHAT IT MUST DO
P0	B	Location pin capture	Ask for a WhatsApp pin, not a typed address. Reverse-geocode, confirm the area name back, price the delivery zone from it.
P0	B	Delivery window	Commit to a window, never a minute. Airport arrivals get a flight-time-aware window.
P0	A	Driver dispatch message	Driver name, phone, car, plate, ETA — sent as a template an hour ahead.
P1	B	Handover photo set	Mandatory: four sides, odometer, fuel gauge, timestamped, attached to the booking. This is the company's entire defence in a damage dispute.
P1	A	Live "driver on the way"	One status ping when the driver departs. Not a tracking map — a message.

During the rental — the part nobody else builds

PRI	AUT	FEATURE	WHAT IT MUST DO
P0	B	Extension	Availability → price difference only → payment → confirm. Minimum 1 day. If the car is taken, offer a substitute rather than refusing.
P0	D	Accident protocol	"Are you safe?" → 999 → don't move the car → police report is mandatory → send location → human in-thread <2 min. Never discuss fault, excess or coverage.
P0	C	Breakdown protocol	Location pin + symptom → roadside dispatch → a human owns the incident to resolution.
P1	B	Salik & fines notification	Proactive, itemised, with date/location/amount/admin fee — as they post, not as a shock at return.
P1	B	Late-return outreach	Message at grace-period start with the hourly rate <i>and the insurance-validity warning</i> . Offer to convert to a formal extension.
P1	B	Running charge total	"What do I owe so far?" answered instantly from the ledger.
P1	C	Upgrade / additional driver	Options + price difference; documents and insurance verified by a human before it takes effect.
P2	A	Mid-rental check-in	One message on day 3 of a long rental: "all good with the car?" Cheap, and catches problems early.

Return & after

PRI	AUT	FEATURE	WHAT IT MUST DO
P0	A	Return reminders	24h and 2h before, with time, place and fuel expectation.
P1	B	Final statement	Itemised close-out: rental, mileage overage, Salik, fines, fuel, cleaning — reconciled against the original quote.
P1	B	Deposit-refund tracker	"Held AED 1,000 · release due by 2 Oct" with proactive updates and a message the day it releases. The single loudest complaint in this market, turned into a feature.
P1	A	Rebook prompt	Timed re-engagement for repeat visitors and monthly renewals.
P2	A	Review request	One ask, only after a clean rental with no dispute.

THE FOUR FEATURES CUSTOMERS WILL ACTUALLY NOTICE

1. An answer in seconds at any hour. 2. A price that doesn't change later. 3. Never being told "sorry, that's booked" without an alternative in the same breath. 4. Being told about their deposit before they have to ask. Everything else is hygiene.

03 The conversation engine

Not a chatbot script. A state machine with a language model attached to the front.

Conversation state — persisted per phone number, forever

```
{
  customer_id, phone, language,           // detected once, switchable, never re-asked
  slots: {
    category | vehicle_id,                 // "an SUV" or a specific unit
    start_at, end_at,                       // ISO, with time - buffers depend on it
    delivery_location { pin | zone_id },
    residency: resident | tourist | gcc,
    driver_age, licence_country,
    extras[], additional_drivers[]
  },
  active_quote: { id, expires_at, total },
  active_hold: { id, vehicle_id, expires_at },
  active_booking_id,
  documents: [{ type, status }],
  escalation: { state, ticket_id, human_id },
  last_intent, unanswered_questions[],    // multi-part messages: nothing gets dropped
  consent: { given_at, purposes[] }
}
```

WHY STATE IS PER-CUSTOMER, NOT PER-CHAT

A tourist rents in March and returns in November. Their name, licence country, language and usual delivery area are already known. **Never ask a returning customer anything they have already told you** — that alone reads as a better company than the competition.

Router — one primary intent, secondary intents preserved

```
NEW_RENTAL · EXISTING_BOOKING · EXTENSION · MODIFICATION · UPGRADE · CANCELLATION
DOCUMENTS · PAYMENT · DELIVERY_STATUS · RETURN · FAQ · DEPOSIT_STATUS
ACCIDENT · BREAKDOWN · FINE_OR_SALIK · COMPLAINT · HUMAN_REQUEST · OTHER
```

RULE	BEHAVIOUR
Accident short-circuit	Any injury/collision signal, in any language, at any point in any flow, jumps straight to the accident protocol. Nothing else is processed first.
Active booking bias	If the number has a live booking, ambiguous messages resolve to <code>EXISTING_BOOKING</code> . "How much for 3 more days?" is an extension, not a new rental.
Multi-intent	Split, queue, answer in order. Unanswered parts stay in <code>unanswered_questions[]</code> until addressed.
Language switch	Switch rendering, keep every slot. Confirm the switch in one line.
Silence	+2h in-window nudge → +24h utility template → release the hold and tell them.

Input handling

INPUT	HANDLING
Voice note	Transcribe → confirm understanding in one line → proceed. Common in Arabic, Hindi, Urdu.
Image	Classify: identity document / damage photo / a car they want (often an Instagram screenshot) / irrelevant. Route accordingly.
Location pin	Reverse-geocode → confirm area name → price the delivery zone.
PDF	Usually a licence or visa. Same pipeline as images.

INPUT	HANDLING
Anything else	Ask once, plainly, what they'd like to do.

Output rules

2–5 lines. One question at a time. Reply buttons for binary choices (max 3), list messages for shortlists (max 10). Money always AED, dates DD MMM YYYY, times 24h Gulf Standard Time — identical in every language. Every message ends with a clear next step.

The happy path, end to end

CUSTOMER

Hi, need a car from tomorrow for 4 days. Something like a Corolla. Deliver to Marina.

`check_availability(compact_sedan, 02-06 Sep, Dubai Marina)`

AGENT

Yes — Toyota Corolla 2024 free for 2–6 Sep, delivered to Dubai Marina.
Quick check: UAE resident or visiting?

CUSTOMER

Visiting, from the UK.

`get_quote(...) → create_hold(...)`

AGENT

A UK licence works here, no IDP needed.

Toyota Corolla 2024 · 2–6 Sep (4 days)

Rental AED 396.00

Delivery Marina AED 0.00

VAT 5% AED 19.80

Total AED 415.80

250 km/day included · AED 1.50 per extra km

Deposit: AED 1,000 card hold, released after return

Not included: Salik (AED 5/gate), fines, fuel

Held until 18:40 today. Lock it in?

[Yes, book it] [Change dates] [Other cars]

CUSTOMER

Yes

AGENT

Two photos please:

1. Passport page + UAE entry stamp
2. UK licence, front and back

`store_documents(...) → create_payment_link(...)`

AGENT

Got them — verifying, usually under 10 minutes.
Here's your payment link: ...

`payment.succeeded webhook → create_booking(...)`

AGENT

Payment received, you're confirmed 

Booking #DXB-10482 · Corolla 2024 · 2–6 Sep

Delivery: 2 Sep, 09:00–10:00, Dubai Marina

Driver details an hour before. Anything else?

Note the ordering: no price before the tool, no "confirmed" before the booking write, no "payment received" before the webhook. That ordering *is* the specification.

04 The tools

The complete surface the model is allowed to touch. It has no database access. Every write is idempotent and fails closed.

```
check_availability({ category?, vehicle_id?, start, end, area? })
  → { units:[{ vehicle_id, make, model, year, class, plate? }],
      alternatives:[...], checked_at }
Failure: returns [] + alternatives. Never throws to the model as "unknown".

get_quote({ vehicle_id|category, start, end, delivery_location, extras[], promo? })
  → { quote_id, line_items[], subtotal, vat, total,
      deposit:{ type:'pre_auth'|'cash'|"none", amount },
      mileage:{ cap_per_day, overage_rate },
      not_included[], exclusions[], expires_at }
The only source of money in the entire system.

create_hold({ quote_id })           → { hold_id, expires_at }
create_booking({ quote_id, customer_id, hold_id, payment_ref, idempotency_key })
  → { booking_id } | { error:'CONFLICT', alternatives[] } | { error:'QUOTE_EXPIRED' }
lookup_bookings({ phone })         → [ booking summaries ]
modify_booking({ booking_id, changes }) → quote | { requires_approval }
extend_booking({ booking_id, new_end }) → quote | { error:'CONFLICT', substitutes[] }
store_documents({ booking_id, media_ids[], type }) → { status:'pending_review', issues[] }
create_payment_link({ quote_id, purpose:'rental'|"deposit"|"extension" })
  → { url, expires_at }
get_charges({ booking_id })
  → { salik[], fines[], mileage_overage, fuel, deposit:{ status, release_due_at } }
schedule_delivery({ booking_id, location, window }) → { driver, eta }
dispatch_roadside({ booking_id, location, symptom }) → { ticket_id, eta }
escalate({ reason, priority, payload }) → { ticket_id }
```

Failure behaviour — specified, not improvised

FAILURE	WHAT THE CUSTOMER SEES
CONFLICT on booking	Honest apology + 2–3 alternatives with prices in the same message. Never "please try again".
QUOTE_EXPIRED	"That price expired — here's the current one" with a fresh quote, not silence.
Tool timeout	One "checking that now" message, then retry. Two failures → escalate. Never invent an answer while waiting.
Payment link fails twice	Offer an alternative method, then escalate to a human.
Any unhandled error	Escalate. The model never sees or repeats a stack trace, tool name or error code.

05 Availability & money

The two things that must be impossible to get wrong.

Availability: make double-booking structurally impossible

Not a check in application code — a constraint the database will not let you violate, including the buffers on both ends.

```
CREATE EXTENSION IF NOT EXISTS btree_gist;

ALTER TABLE bookings ADD CONSTRAINT no_overlap
EXCLUDE USING gist (
  vehicle_id WITH =,
  tstzrange(start_at - prep_buffer, end_at + turnaround_buffer, '[]') WITH &&
) WHERE (status IN ('held','confirmed','active'));
```

Availability is **derived**, never stored. A unit is free for a window only if there is no overlapping row in `bookings`, `maintenance` or `damage_blocks`. Blockers to model:

BLOCKER	TYPICAL SIZE
Turnaround (clean, inspect, refuel)	1–3 hours, per class
Delivery / collection travel	30 min – 3 hours, per zone
Scheduled service	Fleet calendar
Mulkiya (registration) renewal	UAE-specific, annual, blocks the unit
Damage / accident repair	Open-ended until closed
Current renter running late	The most common real-world cause of a broken promise — never assume an on-time return for a same-day handover

HOLD LIFECYCLE

Quote issued → 30-minute hold (longer for rare/exotic units) → expiry stated in the message → auto-release → customer told it was released. A hold that expires silently is worse than no hold.

Money: one object, start to finish

```
quote → booking → final statement
same line-item schema throughout – the invoice is the quote plus actuals

line_items: [
  { code:'RENTAL_DAILY', qty:4, unit:99.00, amount:396.00 },
  { code:'DELIVERY', zone:'marina', amount:0.00 },
  { code:'VAT', rate:0.05, amount:19.80 }
]
deposit: { type:'pre_auth', amount:1000, release_due_at:'2026-10-02' }
mileage: { cap_per_day:250, overage_rate:1.50 }
not_included: ['SALIK','TRAFFIC_FINES','FUEL']
exclusions: ['TYRE_RIM_DAMAGE'] // stated up front, not after the damage
expires_at: '2026-09-01T18:40:00+04:00'
```

THE ACCEPTANCE TEST THAT MATTERS MOST

Quoted total must equal invoiced total, minus only actual usage (Salik, fines, mileage overage, fuel), every single time. Any deviation is a P1 bug, not a rounding discussion. This is the one thing that makes customers trust the whole system.

Payment & deposit — who is allowed to do what

ACTION	OWNER
Explain deposit type and amount	A AI, from the quote
Generate and send a payment link	B Backend → provider. Never a fabricated URL
Create the pre-authorisation hold	B Backend → provider
Confirm payment received	B Provider webhook only
Record cash on delivery	Driver, in the dashboard
Release the hold / refund a deposit	D Finance, named approver. AI reports status only
Any refund or goodwill waiver	D Human, always

Support three payment shapes from day one, because all three exist in this market: **full prepay**, **deposit pre-auth + pay on delivery**, and **pay entirely on delivery**. Pay-on-delivery raises no-show risk — mitigate with a confirmation reply required 2 hours before dispatch.

06 Documents & lifecycle messaging

Document pipeline — WhatsApp is never the filing cabinet

- customer sends photo
 - ▼ webhook downloads media **immediately** (Meta media URLs expire fast)
 - ▼ encrypted private bucket, server-side encryption, signed short-lived URLs
 - ▼ document record: type · expiry · hash · booking_id · status=pending
 - ▼ OCR pre-checks: legible? · name matches booking? · expiry valid? · licence class?
 - ▼ **HUMAN approve / reject with reason** (MVP – never AI-rejected)
 - ▼ booking → documents_verified
 - ▼ conversation log stores a **pointer**, never the image
 - ▼ retention timer: delete after rental + statutory period

	RESIDENT	TOURIST	GCC
Licence	UAE licence mandatory — international licence not accepted	Home-country licence	GCC licence
IDP	n/a	Required if no reciprocal agreement with the UAE	Usually not
ID	Emirates ID, unexpired	Passport, valid through the rental	Passport / GCC ID
Visa	—	Visit visa + entry stamp	—
Card	Credit card in the driver's name for pre-auth (or agreed cash-on-delivery)		

Reciprocal-agreement list must be **data** in the backend, not prompt text — it changes, and getting it wrong either loses a booking or voids insurance.

Message templates to register with Meta

Free-form replies only work inside the 24-hour window after a customer message. Everything else needs an approved *utility* template. Register these on day one:

TEMPLATE	TRIGGER	CONTENTS
booking_confirmed	Booking created	ID, car, dates, address, total, deposit terms
documents_verified	Human approves	Confirmation + what happens next
documents_rejected	Human rejects	Specific reason + what to re-send
delivery_scheduled	Dispatch assigned	Window, driver name, phone, plate
driver_on_the_way	Driver departs	ETA
return_reminder_24h / _2h	Timer	Time, place, fuel expectation
late_return_notice	Grace period ends	Hourly rate + insurance validity warning
charge_posted	Salik / fine posts	Date, location, amount, admin fee
final_statement	Vehicle returned	Itemised close-out vs original quote
deposit_released	Hold released	Amount, date, method
quote_expiring	Hold –10 min	Last chance, one tap to confirm

COST NOTE

Replies inside the customer-initiated 24-hour window are free (plus 1,000 free service conversations/month). Utility templates are cheap. Marketing templates are not — keep them rare and deliberate. Budget for Meta charging in-window service and utility messages from 1 Oct 2026.

07 Human handoff

The measure of a good handoff is that the customer cannot tell one happened — except that it got better.

TRIGGER	PRIORITY	TARGET
Accident or injury	CRITICAL	Human in-thread < 2 min
Breakdown / vehicle immobile	HIGH	< 15 min
Damage, deposit, refund or payment dispute	HIGH	< 15 min
Legal, insurance-claim or police matter	HIGH	< 15 min
Angry or abusive customer (after one de-escalation)	HIGH	< 15 min
Vehicle substitution · VIP · corporate lease · unusual request · low confidence	MEDIUM	< 60 min
"I want to speak to a person"	IMMEDIATE	No questions asked

```
handoff_payload = {
  escalation : { reason, priority, triggered_at, agent_summary }, // one plain sentence
  customer   : { name, phone, language, type, nationality, vip, previous_rentals },
  booking    : { id, vehicle, plate, start, end, delivery_location,
                payment_status, deposit_status, documents_status },
  financial  : { total_paid, outstanding, salik_accrued, fines[] },
  conversation:{ transcript_url, last_10_messages, intents[], sentiment },
  location   : { lat, lng, shared_at },
  attachments: [ handover_photos, police_report_pending ]
}
```

While a human is active the AI is silent. On release it is re-briefed with what the human did, so it never contradicts them. A human can also mark a customer permanently AI-blocked.

ACCIDENT PROTOCOL — THE EXACT SCRIPT

1. "Are you and your passengers safe?" — always first, always alone. 2. If anyone is hurt: call 999, nothing else matters. 3. Don't move the car unless police direct you, or damage is minor with no injuries and it's blocking traffic. 4. A police report is mandatory — insurance won't cover the damage without one. 5. Send your location. 6. Escalate.
- Never discuss fault, excess, coverage, or who pays. Those need a human and a police report.

08 Stack, data & build order

Stack — one choice per layer

LAYER	CHOICE	REASON
Channel	Meta WhatsApp Cloud API (direct)	Flows and interactive messages ship to Cloud first; no BSP margin
Model	Claude Sonnet 5	Multilingual + reliable tool use at chat latency and cost
Orchestration	TypeScript, Anthropic SDK tool loop	Typed tool schemas <i>are</i> the guardrail
API + admin	Next.js on Vercel	Team's stack; webhook and dashboard in one project
Database	Postgres (Neon)	The booking constraint is the product
Async	Vercel Workflow or a Postgres job table	Media pulls and template sends must survive retries
Documents	Vercel Blob / S3, private + SSE	Documents must leave WhatsApp immediately
Payments	Telr or Network International	UAE acquiring, AED settlement, hosted links, pre-auth; Telr bundles Tabby BNPL
Fleet system	The client's existing one. If none: HQ Rental / Appic Fleet / Caryaati	We need an availability source, not a new ERP
Observability	Full tool-call traces per conversation	Every quote must be reconstructable months later

Aviod: n8n/Zapier as the core orchestrator (no transactions or audit trail for money logic) · a vector DB at MVP (the policy KB is small and must be exactly controlled) · fine-tuning (constraints belong in tools) · building a rental ERP.

Data model — the tables you cannot ship without

Customer	phone(unique) · language · residency · nationality · licence_country · consent
Document	type · storage_key · expiry · status · reviewed_by → Customer, Booking
VehicleCategory	min_age · min_licence_years · deposit_default · mileage_cap · overage_rate
Vehicle	plate · vin · mulkiya_expiry · status · odometer · home_zone → VehicleCategory
Booking	times · actual_return · locations · status · quote_id [EXCLUSION CONSTRAINT]
Quote	immutable line_items · total · deposit · expires_at
PricingRule	rate_type · base_rate · season · min_days · zone_fees · validity
Payment / Deposit	provider_ref · amount · type · status · release_due_at · deductions
Delivery / Location	driver · window · status · handover_photos · zone_fee · eta
Maintenance / Damage	downtime_blocks · incident_photos · police_report_ref · liability
Fine / SalikCharge	authority_ref · amount · admin_fee · attributed booking
Conversation / Message	slot state · every message with its tool calls
SupportTicket	reason · priority · payload · assignee · sla_due_at

Build order — and what gates each step

WK	BUILD	GATE BEFORE MOVING ON
1	Discovery: where availability truth lives today; pricing rules; document policy; who takes escalations; WhatsApp number status	We can name the source of truth
1–2	Availability + pricing service against real client data, with the exclusion constraint	A quote is reproducible and correct 100% of the time
3	WhatsApp pipe: Cloud API, webhook, media download, state store, audit log. No AI yet	No message lost or duplicated under retry
4–5	Conversational core: router, slots, FAQ from the KB, EN/AR/RU, tool loop with output guards	Zero invented facts across 200 scripted conversations
5–6	Documents, payment links, atomic booking, confirmation artefact	Zero double-bookings under concurrent load; quote = invoice
6	Delivery, extensions with substitution, accident and breakdown protocols	Safety scripts signed off by operations

WK	BUILD	GATE BEFORE MOVING ON
7	Dashboard: inbox, take-over, escalation queue, document review, pricing editor	Ops can run a full day without a developer
8	Pilot on a traffic subset , human reviewing every conversation	Metrics below, then full launch

THE ONE THING THAT CAN SINK THIS PROJECT

Weeks 1–2. **If availability lives in a spreadsheet or in a manager's head, the AI cannot be truthful** and no amount of prompt engineering fixes it. Do not start the conversation layer until there is a queryable source of truth. If the client has no system, that is the first deliverable — and it is worth building regardless of whether the agent ever ships.

Launch criteria

METRIC	TARGET	WHY
Quote accuracy (quoted = invoiced)	100%	Non-negotiable. Any miss is a P1
Double bookings	0	Enforced by the DB, verified under load
First response time	< 10 s	The best human desks in this market promise 5 minutes
Containment (no human needed)	≥ 60%	The economic case
CRITICAL escalation → human in-thread	< 2 min	Safety
Invented-fact rate in review sample	0	Sample 100 conversations weekly for the first month

C ClientReach .ai

Build spec v1 · 1 September 2026 · Companion to the full research report. Market evidence, competitor analysis and sources live in that document — this one is for building.